

Ch. 4. Multi Variate Review

4.1 Multi Variate Review

- Joint distributions

$$P[(x, y) \in A] = \sum_{x, y \in A} P[(x, y) = (x, y)]$$

$$\sum_{x, y} P[(x, y) = (x, y)] = 1$$

$$F(x, y) = F_{X, Y}(x, y) = P[X \leq x, Y \leq y]$$

$$F_{X, Y}(x, \infty) = F_X(x)$$

$$F_{X, Y}(\infty, y) = F_Y(y)$$

- Marginal Distribution

$$P[X=x] = \sum_y P[(X, Y) = (x, y)]$$

$$P[Y=y] = \sum_x P[(X, Y) = (x, y)]$$

The marginal of X can only depend on x & not y . Likewise, vice-versa.

- Conditional distribution

$$P[Y=y | X=x] = \frac{P[X=x, Y=y]}{P[X=x]}$$

$$= \frac{P[X=x, Y=y]}{\sum_y P[X=x, Y=y]}$$

- X & Y are independent if
 1. the joint distribution factors
 2. support is a rectangle.
- (X, Y) are jointly uniform if all possible (x, y) are equally likely, i.e. the joint PMF is const.
 If (X, Y) are jointly uniform, then conditional distributions are uniform. The marginals need not be uniform.

- Moments

$$E[g(X, Y)] = \sum_{x, y} g(x, y) \cdot P[(X, Y) = (x, y)]$$

$$E[X] = \sum_{x, y} x \cdot P[(X, Y) = (x, y)]$$

$$E[Y^2] = \sum_{x, y} y^2 \cdot P[(X, Y) = (x, y)]$$

$$E[Y | X=x] = E[Y | X] = \sum y \cdot P[Y=y | X=x]$$

$$E[Y^k] = E[E[Y^k | X]]$$

$$\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X])$$

- If X & Y are independent, then $\text{Cov}(X, Y) = 0$.